HARIKRISHNAN K S

Electronics & Communication Engineering Undergraduate | Analog VLSI | Analog IC Design | FPGA
Thrissur, Kerala, India • +91 8590744781 • harikrishnanks.karavattu@gmail.com
Github : harikrishnankskaravattu-eng • **LinkedIn :** linkedin.com/in/harikrishnankstcr/ • **Website :** Portfolio Website

PROFESSIONAL SUMMARY

Electronics and Communication Engineering undergraduate specializing in Analog IC Design. Trained in Analog IC design flow, PDKs, circuit simulation, and Physical Verification (DRC/LVS/ERC/PEX) through industry internship. Strong foundation in semiconductor devices, FPGA design, and Verilog HDL. Seeking to contribute analytical and problem-solving skills to analog circuit design roles.

EDUCATION

B.Tech, Electronics & Communication Engineering *Expected 2027*
Government Engineering College, Sreekrishnapuram, Palakkad
APJ Abdul Kalam Technological University, Kerala
CGPA : **8.05** *(up to semester 6)*

TECHNICAL SKILLS

Languages:	C/C++, Python, Verilog HDL
Analog VLSI:	**Analog IC Design Flow** (Fundamentals), Analog Layout Basics, Concepts of PDKs, Semiconductor Devices, **Physical Verification Concepts** (DRC, LVS, ERC), Parasitic Extraction concepts (PEX), Corner Analysis basics
FPGA & Tools:	**Vivado 2022.2**, Icarus Verilog, GTKWave, Zynq-7000 (ZedBoard), FSM, RTL, Synthesis, STA, Testbench Design
Embedded & Hardware:	ESP32, STM32, Arduino, UART, I2C, SPI
Software:	**LTSpice**, Proteus, TinkerCad, Figma

TECHNICAL PROJECTS

Design and Layout of CMOS Bandgap Reference Circuit | SKY130 PDK • Xschem • Magic • Ngspice *(Ongoing)*

- Designing and simulating a CMOS bandgap reference circuit in Xschem using the SKY130 open-source PDK, targeting stable, temperature-independent voltage reference output.
- Performing schematic capture in Xschem and full-custom layout in Magic, applying device matching and common-centroid layout techniques for improved accuracy and physical verification (DRC/LVS) readiness.

Adaptive Real-Time Traffic Light Control Using FPGA | Zedboard • FSM • Verilog • Vivado

- Developed FPGA-based traffic control system in Verilog HDL, improving traffic flow by 25% on Zynq-7000 (ZedBoard) using FSM-based control logic.
- Simulated and verified system behavior using GTKWave and Icarus Verilog

Enviroguard – Environmental Monitoring System | STM32 • IoT • Proteus

- Designed STM32-based environmental monitoring system, improving power efficiency by 30% with real-time sensor alerts (latency <50 ms).
- Validated circuit design in Proteus prior to hardware implementation.

RFID-Based Attendance System | ESP8266 • IoT

- Developed RFID-based attendance system using ESP8266, enabling 100% contactless tracking and real-time data processing.

INTERNSHIP EXPERIENCE

Analog IC Design Trainee Intern *June 2026*
Sophrosyne Semiconductor, Kunjibettu, Udupi

- Studied complete Analog IC Design Flow — schematic design, layout, physical verification (DRC/LVS/ERC/PEX) — using PDKs, technology libraries, and analog building blocks (MOSFETs, current mirrors, differential amplifiers, resistors, capacitors, diodes).
- Gained hands-on understanding of device matching, common-centroid layout, guard rings, corner analysis, Monte Carlo analysis, and reliability concepts for robust analog design.

WORKSHOPS & CERTIFICATIONS

Advanced Driver Assistance Systems & Machine Learning Workshop ***November 2023***
Indian Institute of Technology Delhi
Introduction to Internet of Things **November 2024**
NPTEL

ACHIEVEMENTS & LEADERSHIPS

- Active member, Robotics Club, GEC Sreekrishnapuram; Volunteer, IEEE Young Buddies 2.0; UI/UX Designer & Design Head, ECE Department (incl. department magazine).